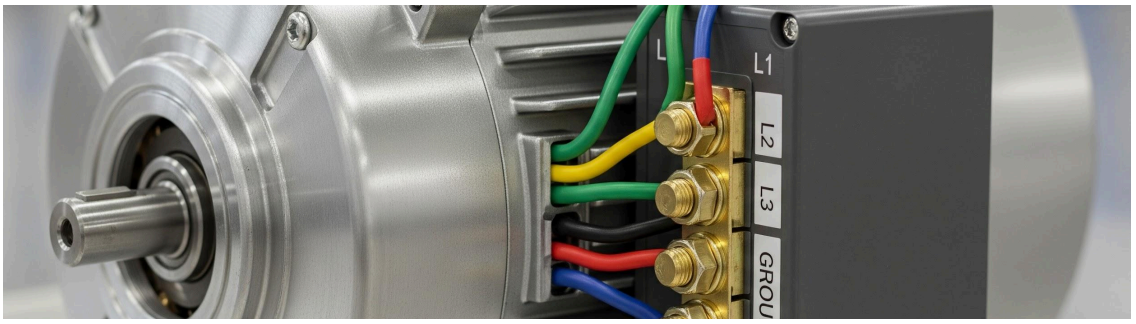

Electrical Machine Data Sheet

Machine Name/Type : Three Phase Slip Ring Induction Motor

Name plate Details

The rating section details the machine's operational capacity and design limits.

PARAMETER	VALUE	UNIT
Power output	5	HP
Stator Voltage	415	V
Stator Current	7.5	A
Rotor Voltage	200	A
Rotor current	11	A
Frequency	50	Hz
Speed	1410	RPM

Precautions

I. Ensure the supply is **415 V, 3-phase AC** before starting.

II. Check correct **phase sequence** before operation.

III. Ensure external rotor resistance is **fully inserted at start**.

IV. Gradually cut out rotor resistance as the motor picks up speed.

V. Never short the slip rings at starting condition.

VI. Ensure proper brush contact on slip rings.

VII. Maintain proper earthing of the motor body.

VIII. Do not overload the motor beyond rated capacity.

IX. Monitor current, temperature, and speed continuously.

X. Switch OFF immediately if excessive sparking, vibration, or overheating occurs.

Working Principle of Three Phase Slip Ring Induction Motor

When a three-phase AC supply is applied to the stator, it produces a **rotating magnetic field** at synchronous speed.

This rotating field induces current in the **wound rotor**, which is connected to external resistances through slip rings.

The added rotor resistance improves starting torque and limits starting current.

As the motor accelerates, the external resistance is gradually reduced, and the motor runs close to synchronous speed with minimum slip.

Working Characteristics

I. **High starting torque** compared to squirrel cage motor.

II. **Lower starting current** due to external rotor resistance.

III. Smooth and controlled acceleration.

IV. Speed regulation poorer than squirrel cage motor.

V. Higher maintenance due to brushes and slip rings.

VI. Suitable for heavy starting loads.

Applications

1) Cranes
2) Hoists
3) Elevators
4) Crushers
5) Rolling mills
6) Conveyors
7) Mine winders
8) Compressors
9) Heavy-duty pumps
10) Industrial lifting equipment